

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							*****VALVES*****			810	NIPPLE TO BE CUT FROM SCH 160 PIPE.
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
20							***** MODULAR VALVES *****				
30										817	USE ONLY FOR TIE IN
30	U04		MODULAR DBB	K15AM10	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	820	ALLOY 825 INTERNAL CLADDING IS ACCEPTABLE FOR PIPES MANUFACTURED FROM METALLURGICALLY ROLL-BONDED OR EXPLOSION BONDED CLAD PLATES. ALLOY 625 MATERIAL TO BE USED FOR PIPES MANUFACTURED WITH INTERNAL WELD OVERLAY.
40			MODULAR DBB	K15AM14	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		
50		VAV	MODULAR DBB	K09AM14	3	3	MODULAR VALVE DB&B BALL TYPE 900# RF INC 825 INC 625 TRIM METAL LO FULL SIZE TYPE 2BV	0	0		
60			MODULAR DBB	M15AM10	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0		
70			MODULAR DBB	M15AM14	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0		
80		VAW	MODULAR DBB	M15EM10	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	B1	ALL BOLTS ARE METRIC DIA.
90										B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
90	U04	PS4	PIPE	5358507	3	3	PIPE SCH 80 SMLS LTCS A333 GR.6 BE	0	0		
100											
100		VAW	MODULAR DBB	M09EM10	1	1	MODULAR VALVE DB&B BALL TYPE 900# RF INC 825 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	B5	7/8" X 155MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AC02 WHEREEVER MATCHED WITH A SPECTACLE BLIND
110							***** BALL VALVES *****			B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
110		VB2									
120			BALL	A15AM13	2	2	BALL 1500# RF 825 INCONEL 625 TRIM TRUNNION SOFT RB LO	0	0		
130			BALL	A09AM13	3	3	BALL 900# RF 825 INCONEL 625 TRIM TRUNNION SOFT RB LO	0	0		
140			BALL	A09AR15	4	10	BALL 900# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT RB GO	0	0	BA4	TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.
150	U04	817	BALL	A09CR15	12	16	BALL 900# CF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT RB GO	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
160		VAA, 817	BALL	A09CR16	30	30	BALL 900# CF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION METAL RB GO	0	0		
170			BALL	B15AM09	0.75	1.5	BALL 1500# RF 825 INCONEL 625 TRIM FLOATING SOFT FB LO	0	0	FF5	FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES
180		VA5	BALL	B15AM13	2	2	BALL 1500# RF 825 INCONEL 625 TRIM TRUNNION SOFT FB LO	0	0		
190	U04	VA5	BALL	B09AM15	3	3	BALL 900# RF 825 INC 625 TRIM TRUNNION SOFT FB GO	0	0	GA6	SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117
200		VA5	BALL	B09AR15	4	10	BALL 900# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT FB GO	0	0		
210	U04	817	BALL	B09CR15	12	16	BALL 900# CF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT FB GO	0	0	ND9	USE ONLY WITH CONSENT OF MATERIALS ENGINEER.
220	U04		BALL	B15AM10	0.75	1.5	BALL 1500# RF 825 INC 625 IN TRIM FLOATING METAL FB LO	0	0		
230	U04		BALL	A15AM14	2	2	BALL 1500# RF 825 INC 625 TRIM TRUNNION METAL RB LO	0	0	NT6	THIS ITEM IS SUBJECT TO CONFIRMATION BY VENDOR DATA.
240	U04		BALL	A09AR16	4	10	BALL 900# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION METAL RB GO	0	0		
250							***** GATE VALVES *****			PS1	PIPE ONLY TO BE USED AS TRUNNION SUPPORTS
250		VQ2									
260			GATE	G15AM49	0.75	1.5	GATE 1500# RF 825 INC 625 INC TRIM WEDGE HW	0	0	PS4	PIPE ONLY TO BE USED AS TRUNNION SUPPORTS. USE FOR HEADER SIZE 4" AND ABOVE ONLY.
270							***** GLOBE VALVES *****			PS5	PIPE ONLY TO BE USED FOR RPAD MATERIAL
270										SB5	USE JACK SCREWS ONLY WHERE PIPING CAN NOT BE OPENED WITH FLANGE SPREADER.
280			GLOBE	L15AM43	0.75	2	GLOBE 1500# RF INC 825 INC TRIM 625 OS&Y HW	0	0		
290			GLOBE	L09AM43	3	3	GLOBE 900# RF INC 825 INC TRIM 625 OS&Y HW	0	0	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
300			GLOBE	L09AR44	4	8	GLOBE 900# RF LTCS A352 LCC + 825 CLAD INC 625 TRIM OS&Y GO	0	0		
310							***** CHECK VALVES *****			UI2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
310											
320		VC1	CHECK	C15AM51	0.75	1.5	CHECK PISTON 1500# RF INC 825 TRIM 625	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
330			CHECK	C15AM46	2	2	CHECK DUAL WAFER LUG 1500# RF INC 825 TRIM 625 THROU BOLTED	0	0		
340			CHECK	C09AM46	3	3	CHECK DUAL WAFER LUG 900# RF INC 825 TRIM 625 THROU BOLTED	0	0		
350			CHECK	C09AR46	4	8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC + 825 CLAD TRIM INC 625 THROU BOLTED	0	0	VAA	METAL SEATED VALVE USED DUE TO MANUFACTURING LIMITATIONS OF SOFT SEATED VALVE.
360			CHECK	C09AR53	10	10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC + 825 CLAD TRIM 625	0	0		
370	U04	817	CHECK	C09CR53	12	16	CHECK DUAL DBL FLG 900# CF LTCS A352 LCC + 825 CLAD INC 625 TRIM	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
380							*****PIPES*****			VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
380											
390											
390			PIPE	5732001	0.75	1.5	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 PE	0	0		
400			PIPE	5361607	2	3	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 BE	0	0		
410		810	PIPE	5732003	0.75	0.75	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 PE	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
420	U04	820	PIPE	5358521_C60	18	24	PIPE SCH 100 EFW LTCS A671 CC60 CL.22 BE WITH 3MM INC 825/625 INTERNAL CLADDING	0	0		
430	U04		PIPE	5358509_C60	8	16	PIPE SCH 100 SMLS LTCS A333 GR.6 BE WITH 3MM INC 625 INTERNAL CLADDING	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
440	U04		PIPE	5358510_C60	4	6	PIPE SCH 120 SMLS LTCS A333 GR.6 BE WITH 3MM INC 625 INTERNAL CLADDING	0	0		
450	U04	820	PIPE	5358635_C60	30	30	PIPE WT 48.0MM EFW LTCS A671 CC60 CL.22 BE WITH 3 MM INC 825/625 INTERNAL CLADDING	0	0	VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.
460	U04	820	PIPE	5358533_C60	36	36	PIPE WT 57.0MM EFW LTCS A671 CC60 CL.22 BE WITH 3 MM INC 825/625 INTERNAL CLADDING	0	0		
470							***** TRUNNION PIPE SUPPORTS *****				
470											
480											
480	U04	PS1	PIPE	5358521	20	24	PIPE SCH 100 EFW LTCS A671 CC60 CL.22 BE	0	0		
490	U04	PS1	PIPE	5358509	8	16	PIPE SCH 100 SMLS LTCS A333 GR.6 BE	0	0		
500	U04	PS1	PIPE	5358510	4	6	PIPE SCH 120 SMLS LTCS A333 GR.6 BE	0	0		
510	U04	PS5	PIPE	5358533	36	36	PIPE WT 57.0MM EFW LTCS A671 CC60 CL.22 BE	0	0		
520											

	TENGIZCHEVROI TCO, TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES BUTT - WELD PWHT FOR SIZE 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 220 °C	12/16/2024	BY OZAC	U04	Page 1 of 4	900C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
520							***** FITTINGS *****				
530			ELBOW	5425111	0.75	3	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825	0	0		
540			ELBOW	5392700_C80	4	6	ELL 90 DEG LR SCH 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	0	0		
550			ELBOW	5392654_C80	8	16	ELL 90 DEG LR SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	0	0		
560			ELBOW	5392605_C80	18	24	ELL BW 90 DEG LR SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	0	0		
570			ELBOW	5392563_C80	30	30	ELL 90 DEG LR WT 48.0MM LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	0	0		
580			ELBOW	5392564_C80	36	36	ELL 90 DEG LR WT 57.0MM LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	0	0		
590	U04		ELBOW	5392654_C60	8	16	ELL 90 DEG LR SCH 100 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0		
600	U04		ELBOW	5392563_C60	30	30	ELL 90 DEG LR WT 48.0MM LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	0	0		
610	U04		ELBOW	5392564_C60	36	36	ELL 90 DEG LR WT 57.0MM LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	0	0		
620	U04		ELBOW	5392605_C60	18	24	ELL BW 90 DEG LR SCH 100 LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	0	0		
630	U04		ELBOW	5392700_C60	4	6	ELL BW 90 DEG LR SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0		
640			ELBOW	5425113	0.75	3	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825	0	0		
650			ELBOW	5392701_C80	4	6	ELL BW 45 DEG LR SCH 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	0	0		
660			ELBOW	5392660_C80	8	16	ELL 45 DEG LR SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	0	0		
670			ELBOW	5392638_C80	36	36	ELL 45 DEG LR WT 57.0MM LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	0	0		
680			ELBOW	5392604_C80	18	24	ELL 45 DEG LR SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	0	0		
690	U04		ELBOW	5392660_C60	8	16	ELL 45 DEG LR SCH 100 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0		
700	U04		ELBOW	5392604_C60	18	24	ELL 45 DEG LR SCH 100 LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	0	0		
710	U04		ELBOW	5392638_C60	36	36	ELL 45 DEG LR WT 57.0MM LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	0	0		
720	U04		ELBOW	5392701_C60	4	6	ELL BW 45 DEG LR SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0		
730											
730			TEE	5425119	0.75	3	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825	0	0		
740			TEE	5392702_C80	4	6	TEE BW SCH 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	0	0		
750			TEE	5392663_C80	8	16	TEE SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	0	0		
760			TEE	5392602_C80	18	24	TEE SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	0	0		
770			TEE	5392639_C80	36	36	TEE BW WT 57.0MM LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	0	0		
780	U04		TEE	5392702_C60	4	6	TEE BW SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0		
790	U04		TEE	5392639_C60	36	36	TEE BW WT 57.0MM LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	0	0		
800			TEE RED	5425126	0.75	2	TEE RED SCH 80S ALLOY 825 B366-WPS UNS N08825	1	3		
810			TEE RED	5392620	3	3	TEE RED SCH 120/80S ALLOY 825 B366-WPS UNS N08825	4	6		
820			TEE RED	5392622_C80	4	6	TEE RED SCH 100/ 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	8	16		
830			TEE RED	5392662_C80	8	14	TEE RED SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	10	16		
840			TEE RED	5392621_C80	8	20	TEE RED SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	18	24		
850		NT6	TEE RED	5392836_C80	14	14	TEE RED WT 57.0MM/SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	36	36		
860	U04		TEE RED	5392662_C60	8	14	TEE RED SCH 100 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	10	16		
870	U04		TEE RED	5392621_C60	8	20	TEE RED SCH 100 LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	18	24		
880	U04		TEE RED	5392622_C60	4	6	TEE RED SCH 100/ 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	8	16		
890	U04		TEE RED	5392603_C60	4	4	TEE RED SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	6	6		
900	U04		TEE RED	5392471_C60	3	3	TEE RED SCH 120/160 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	4	6		
910	U04	NT6	TEE RED	5392836_C60	14	14	TEE RED WT 57.0MM/SCH 100 LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	36	36		
920	U04		TEE RED	5392836_C60	24	24	TEE RED WT 57.0MM/SCH 100 LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	36	36		
930	U04		TEE	5392663_C60	8	16	TEE SCH 100 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0		
940	U04		TEE	5392602_C60	18	24	TEE SCH 100 LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	0	0		
950											
950			CAP	5424270	0.75	3	CAP SCH 80S ALLOY 825 B366 UNS N08825	0	0		
960			CAP	5392703_C80	4	6	CAP SCH 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	0	0		
970			CAP	5392664_C80	8	24	CAP SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	0	0		
980	U04		CAP	5392664_C60	8	24	CAP SCH 100 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0		
990	U04		CAP	5392703_C60	4	6	CAP SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0		
1000											
1000			RED CONC	5425122	0.75	2	REDUCER CONC SCH 80S ALLOY 825 B366-WPS N08825	1	3		
1010			RED CONC	5392720	3	3	REDUCER CONC SCH 120/80S ALLOY 825 B366-WPS N08825	4	6		
1020			RED CONC	5392722_C80	4	6	REDUCER CONC SCH 100/ 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	8	14		
1030			RED CONC	5392665_C80	8	14	REDUCER CONC SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	10	16		
1040			RED CONC	5392721_C80	10	20	REDUCER CONC SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	18	24		
1050	U04		RED CONC	5392665_C60	8	14	REDUCER CONC SCH 100 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	10	16		
1060	U04		RED CONC	5392721_C60	10	20	REDUCER CONC SCH 100 LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	18	24		
1070	U04		RED CONC	5392722_C60	4	6	REDUCER CONC SCH 100/ 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	8	14		
1080	U04		RED CONC	5392775	2	3	REDUCER CONC SCH 160/80S ALLOY 825 B366-WPS N08825	4	6		
1090			RED ECC	5425123	0.75	2	REDUCER ECC SCH 80S ALLOY 825 B366-WPS N08825	1	3		
1100			RED ECC	5392723	3	3	REDUCER ECC SCH 120/80S ALLOY 825 B366-WPS N08825	4	6		
1110			RED ECC	5392725_C80	4	6	REDUCER ECC SCH 100/ 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	8	14		
1120			RED ECC	5392666_C80	8	14	REDUCER ECC SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	10	16		
1130			RED ECC	5392724_C80	10	20	REDUCER ECC SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	18	24		
1140			RED ECC	5392213_C80	30	30	REDUCER ECC WT 57.0MM/48.0MM LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	36	36		
1150	U04		RED ECC	5392666_C60	8	14	REDUCER ECC SCH 100 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	10	16		
1160	U04		RED ECC	5392724_C60	10	20	REDUCER ECC SCH 100 LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	18	24		
1170	U04		RED ECC	5392725_C60	4	6	REDUCER ECC SCH 100/ 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	8	14		
1180	U04		RED ECC	5392705_C60	4	4	REDUCER ECC SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	6	6		
1190	U04		RED ECC	5392776	2	3	REDUCER ECC SCH 160/80S ALLOY 825 B366-WPS N08825	4	6		
1200	U04		RED ECC	5392213_C60	30	30	REDUCER ECC WT 57.0MM/48.0MM LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	36	36		

	TENGINZCHEVROIL TCO KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES BUTT - WELD PWHT FOR SIZE 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 220 °C	12/16/2024	BY OZAC	U04	Page 2 of 4	900C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
1210											
1210			OLETS	5514631	0.75	1	NIPOFLANGE 1500# RF SCH 80S ALLOY 825 B564 UNS N08825	2	3		
1220			OLETS	5513421	0.75	2	NIPOFLANGE 1500# RF SCH 120/80S ALLOY 825 B564 UNS N08825	4	6		
1230			OLETS	5513422	0.75	2	NIPOFLANGE 1500# RF SCH 100/80S ALLOY 825 B564 UNS N08825	8	24		
1240			OLETS	5516678	0.75	2	NIPOFLANGE 1500# RF WT 48.0MM /80S ALLOY 825 B564 UNS N08825	30	30		
1250			OLETS	5516679	2	2	NIPOFLANGE 1500# RF WT 57.0MM /80S ALLOY 825 B564 UNS N08825	36	36		
1260		UJ2	OLETS	5513423	1.5	1.5	NIPOFLANGE 1500# RF SCH 120/160 ALLOY 825 B564 UNS N08825	4	6		
1270		UJ2	OLETS	5513424	1.5	1.5	NIPOFLANGE 1500# RF SCH 100/160 ALLOY 825 B564 UNS N08825	8	24		
1280		FF5	OLETS	5504591	0.75	1.5	NIPOFLANGE FLAT BOTTOM 1500# RF SCH 80S ALLOY 825 B564 UNS N08825	0	0		
1290	U04	UJ2	OLETS	5513426	1.5	1.5	NIPOFLANGE 1500# RF 48MM/SCH 160 ALLOY 825 B564 UNS N08825	30	30		
1300	U04	UJ2	OLETS	5513425	1.5	1.5	NIPOFLANGE 1500# RF 57.00MM/SCH 160 ALLOY 825 B564 UNS N08825	36	36		
1310											
1310			OLETS	5513414	0.75	1	WELDOLET SCH 80S ALLOY 825 B564 UNS N08825	2	3		
1320			OLETS	5509620	0.75	3	WELDOLET SCH 120 /80S ALLOY 825 B564 UNS N08825	4	6		
1330			OLETS	5509621	0.75	3	WELDOLET SCH 100 /80S ALLOY 825 B564 UNS N08825	8	24		
1340			OLETS	5505645	0.75	3	WELDOLET WT 48.0MM/SCH 80S ALLOY 825 B564 UNS N08825	30	30		
1350	U04		OLETS	5505344_C60	8	8	WELDOLET SCH 100 LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	18	24		
1360	U04		OLETS	5509622_C60	4	6	WELDOLET SCH 100/120 LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	18	24		
1370	U04		OLETS	5505382_C60	14	14	WELDOLET WT 57.0MM / SCH 100 LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	36	36		
1380	U04		OLETS	5509622_C80	4	6	WELDOLET SCH 100/120 LTCS A350 LF2 CL.1 WITH 3MM INC 825 INTERNAL CLADDING	18	24		
1390	U04		OLETS	5505344_C80	8	8	WELDOLET SCH 100 LTCS A350 LF2 CL.1 WITH 3MM INC 825 INTERNAL CLADDING	18	24		
1400											
1400		UJ2	ELBOFLG	5516020	1.5	1.5	ELBOFLANGE 1500# RF SCH 120/160 ALLOY 825 B564 UNS N08825	4	6		
1410		UJ2	ELBOFLG	5516022	1.5	1.5	ELBOFLANGE 1500# RF SCH 100/160 ALLOY 825 B564 UNS N08825	8	24		
1420											
1420		816					***** FLANGES *****				
1430											
1430		FA8	FLANGE	5614400	0.5	0.5	FLG SCRDR (FNPT) REDUCING 1500# RF ALLOY 825 B564 UNS N08825	0.75	1.5		
1440											
1440			FLANGE	5851205	0.75	0.75	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0		
1450			FLANGE	5851205	2	2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0		
1460	U04	ND9	FLG ADAPTER	5851221	0.75	0.75	FLANGE ADAPTER 1500# RF ALLOY 825 B564 UNS N08825 X 12 MM TUBE	0	0		
1470	U04	ND9	FLG ADAPTER	5851221	2	2	FLANGE ADAPTER 1500# RF ALLOY 825 B564 UNS N08825 X 12 MM TUBE	0	0		
1480											
1480			FLANGE	5570286	0.75	2	FLG WN 1500# RF ALLOY 825 SCH 80S B564 UNS N08825	0	0		
1490			FLANGE	5557401	3	3	FLG WN 900# RF ALLOY 825 SCH 80S B564 UNS N08825	0	0		
1500	U04		FLANGE	5557143_C80	4	6	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 120 WITH 3MM INC 825 CLAD FACE & BORE	0	0		
1510	U04		FLANGE	5557142_C80	8	10	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 100 WITH 3MM INC 825 CLAD FACE & BORE	0	0		
1520	U04		FLANGE	5557142_C80	12	24	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 100 WITH 3MM INC 825 CLAD FACE & BORE	0	0		
1530	U04		FLANGE	5557142_C60	8	10	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 100 WITH 3MM INC 625 CLAD FACE & BORE	0	0		
1540	U04		FLANGE	5557142_C60	12	24	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 100 WITH 3MM INC 625 CLAD FACE & BORE	0	0		
1550	U04		FLANGE	5557143_C60	4	6	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 120 WITH 3MM INC 625 CLAD FACE & BORE	0	0		
1560	U04	ND9		5557148_C60	30	30	FLG WN 900# RF LTCS A350 LF2 CL.1 WT 48.0MM B16.47 SERIES A WITH 3 MM INC 625 CLAD FACE & BORE	0	0		
1570	U04		FLANGE	5557149_C60	36	36	FLG WN 900# RF LTCS A350 LF2 CL.1 WT 57.0MM B16.47 SERIES A WITH 3 MM INC 625 CLAD FACE & BORE	0	0		
1580	U04	817	FLANGE	5619887_C80	12	24	FLG COMPACT 900# RJ LTCS A350 LF2 CL.1 SCH 100 WITH 3MM INC 825 CLAD FACE & BORE	0	0		
1590	U04	817	FLANGE	5619888_C80	30	30	FLG COMPACT 900# RJ LTCS A350 LF2 CL.1 WT 48.0MM WITH 3MM INC 825 CLAD FACE & BORE	0	0		
1600	U04	817	FLANGE	5619889_C80	36	36	FLG COMPACT 900# RJ LTCS A350 LF2 CL.1 WT 57.0MM WITH 3MM INC 825 CLAD FACE & BORE	0	0		
1610	U04	817	FLANGE	5619888_C60	30	30	FLG COMPACT 900# RJ LTCS A350 LF2 CL.1 WT 48.0MM WITH 3MM INC 625 CLAD FACE & BORE LINE CLASS 900C80	0	0		
1620	U04	817	FLANGE	5619889_C60	36	36	FLG COMPACT 900# RJ LTCS A350 LF2 CL.1 WT 57.0MM WITH 3MM INC 625 CLAD FACE & BORE LINE CLASS 900C80	0	0		
1630			BLIND	5617185	0.5	2	FLG BLIND 1500# RF ALLOY 825 B564 UNS N08825	0	0		
1640			BLIND	5617190	3	3	FLG BLIND 900# RF ALLOY 825 B564 UNS N08825	0	0		
1650	U04		BLIND	5615605_C80	4	10	FLG BLIND 900# RF LTCS A350 LF2 CL.1 WITH 3 MM INC 825 CLAD FACE	0	0		
1660	U04		BLIND	5615605_C80	12	24	FLG BLIND 900# RF LTCS A350 LF2 CL.1 WITH 3 MM INC 825 CLAD FACE	0	0		
1670	U04		BLIND	5615605_C60	4	10	FLG BLIND 900# RF LTCS A350 LF2 CL.1 WITH 3 MM INC 625 CLAD FACE	0	0		
1680	U04		BLIND	5615605_C60	12	24	FLG BLIND 900# RF LTCS A350 LF2 CL.1 WITH 3 MM INC 625 CLAD FACE	0	0		
1690	U04	817	FLANGE	5619887_C60	12	24	FLG COMPACT 900# RJ LTCS A350 LF2 CL.1 SCH 100 WITH 3MM INC 625 CLAD FACE & BORE LINE CLASS 900C80	0	0		
1700	U04	817	BLIND	5619726	12	16	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 CL.1 WITH 3 MM INC 825 CLAD FACE PIPE CLASS 900C80	0	0		
1710	U04	817	BLIND	5619726	30	36	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 CL.1 WITH 3 MM INC 825 CLAD FACE PIPE CLASS 900C80	0	0		
1720	U04	817	BLIND	5619726_C60	12	16	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 CL.1 WITH 3 MM INC 625 CLAD FACE LINE CLASS 900C80	0	0		
1730	U04	817	BLIND	5619726_C60	30	36	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 CL.1 WITH 3 MM INC 625 CLAD FACE LINE CLASS 900C80	0	0		
1740											
1740							***** ORIFICE FLANGES *****				
1750											
1750			ORIFICE FLG	5651001	2	2	WN ORIF SET 1500# RF ALLOY 825 B564 UNS N08825 SCH 80S	0	0		
1760			ORIFICE FLG	5650941	3	3	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 80S	0	0		
1770			ORIFICE FLG	5624532_C80	4	6	WN ORIF SET 900# RF LTCS A350 SCH 120 WITH 3MM INC 825 CLAD FACE & BORE	0	0		
1780			ORIFICE FLG	5624531_C80	8	24	WN ORIF SET 900# RF LTCS A350 SCH 100 WITH 3MM INC 825 CLAD FACE & BORE	0	0		
1790	U04		ORIFICE FLG	5624531_C60	8	24	WN ORIF SET 900# RF LTCS A350 SCH 100 WITH 3MM INC 625 CLAD FACE & BORE	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES BUTT - WELD PWHT FOR SIZE 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 220 °C	12/16/2024	BY OZAC	U04	Page 3 of 4	900C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1800	U04		ORIFICE FLG	5624532_C60	4	6	WN ORIF SET 900# RF LTCS A350 SCH 120 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
1810										
1810							***** PIPE NIPPLES *****			
1820										
1820		810		0			NIPPLE PBE TO BE CUT FROM PIPE			
1830										
1830							***** GASKETS *****			
1840										
1840			GASKET	5673800	0.5	2	GASKET 1500# ALLOY 825 SPIRAL WOUND GRAPHITE IR	0	0	
1850			GASKET	5673700	3	24	GASKET 900# ALLOY 825 SPIRAL WOUND GRAPHITE IR	0	0	
1860	U04		GASKET	5674035	30	36	GASKET 900# ALLOY 825 KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
1870	U04	817	GASKET	5673431	12	36	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 900C80	0	0	
1880	U04	GA6	GASKET	5673023	2	2	GASKET 1500# ALLOY 825 KAMMPROFILE GRAPHITE ASME B16.5	0	0	
1890										
1890		B1					***** BOLTING *****			
1900										
1900			BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1910		BA4	BOLTING	5675776	0.5	99	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
1920										
1920		SB6	BOLTING	5675675	10	24	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1930										
1930		SB5					***** BLINDS & SPACERS *****			
1940										
1940			SPEC BLIND	5750300	0.75	2	SPECTACLE BLIND 1500# RF ALLOY 825 B424 UNS N08825	0	0	
1950			SPEC BLIND	5750400	3	3	SPECTACLE BLIND 900# RF ALLOY 825 B424 UNS N08825	0	0	
1960			SPEC BLIND	5749596_C80	4	8	SPECTACLE BLIND 900# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE	0	0	
1970	U04		SPEC BLIND	5749596_C60	4	8	SPECTACLE BLIND 900# RF LTCS A516 GR.70 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
1980										
1980			PADDLE	5750102_C80	10	10	PADDLE BLANK 900# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE	0	0	
1990			PADDLE	5750102_C80	12	24	PADDLE BLANK 900# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE	0	0	
2000			PADDLE	5750202_C80	10	10	PADDLE SPACER 900# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2010			PADDLE	5750202_C80	12	24	PADDLE SPACER 900# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2020	U04	817	PADDLE	5750264_C80	12	24	PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE	0	0	
2030	U04	817	PADDLE	5750265_C80	12	24	PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2040	U04		PADDLE	5750102_C60	10	10	PADDLE BLANK 900# RF LTCS A516 GR.70 WITH 3MM INC 625 CLAD FACE	0	0	
2050	U04		PADDLE	5750102_C60	12	24	PADDLE BLANK 900# RF LTCS A516 GR.70 WITH 3MM INC 625 CLAD FACE	0	0	
2060	U04	817	PADDLE	5750264_C60	12	24	PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70 WITH 3MM INC 625 CLAD FACE	0	0	
2070	U04		PADDLE	5750202_C60	10	10	PADDLE SPACER 900# RF LTCS A516 GR.70 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2080	U04		PADDLE	5750202_C60	12	24	PADDLE SPACER 900# RF LTCS A516 GR.70 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2090	U04	817	PADDLE	5750265_C60	12	24	PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2100										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
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